

NeoMundi × Oak & Sparrow

Contribution to Common Metrological Layer for Heterogeneous AI Infrastructures

Measure. Determine. Enforce.

Oak & Sparrow Systems Enterprise | Research contribution for NeoMundi

Executive abstract

Oak & Sparrow Systems Enterprise integrated NeoMundi's runtime measurement layer into a pre-existing deterministic policy, action-control, and evidence architecture. The experiment tested whether independent behavioral measurements could be consumed without transferring interpretation or authorization authority to the measurement layer. Across a focused 60-request experiment and a controlled 1,000-request benchmark, the articulation preserved a clear sequence: NeoMundi measured, OASSE interpreted the measurements under declared policy, and Gatekeeper produced and sealed the resulting operational action. The evidence supports complementary operational value in routing, traceability, auditability, and evidence quality. It provides one substantive indication in support of a common metrological layer, while leaving cross-architectural generality as an open comparative question.

1. Infrastructure and original problem

Oak & Sparrow Systems Enterprise develops Gatekeeper, a proprietary deterministic security, policy-determination, action-control, and evidence architecture for AI-enabled and automated systems.

Gatekeeper treats a proposed machine action as a request for authority rather than permission to execute. It is designed to evaluate the request under declared policy, determine the applicable outcome, apply the resulting control, and preserve verifiable evidence of the decision.

The NeoMundi articulation focused on OASSE's FAM family and guardian governance surface. The experimental question was narrow and testable: can an independent runtime measurement layer be integrated into a deterministic policy and enforcement architecture while preserving a clear separation between measurement, interpretation, and final action?

2. Components that existed before NeoMundi

Before the articulation, OASSE already had:

- a sealed ontology and deterministic policy packs
- fail-closed signal extraction
- jurisdiction and age-band applicability logic
- deterministic rule evaluation and verdict precedence
- FAM policy determinations and an escalating hold ladder
- policy and statutory provenance
- sealed decision artifacts and SHA-256-linked evidence persistence
- an HTTP service and browser-based governance console
- controlled test corpora and durable benchmark runners

NeoMundi did not create OASSE's policy authority, FAM determinations, Gatekeeper actions, or evidence-chain mechanism.

3. Integration method

NeoMundi was integrated through its deployed runtime API using an OASSE-built bridge. The validated V1 path was:

1. Load a controlled request and its NeoMundi measurement payload.
2. Send the payload through NeoMundi's deployed OBS interface.
3. Preserve and normalize the NeoMundi response.
4. Convert selected measurements into versioned and traceable OASSE evidence facts.
5. Evaluate the original request independently through the FAM policy engine.
6. Select the strictest combined thermal state.
7. Produce a Gatekeeper action of PROCEED, HOLD, or DENY.
8. Seal the composite evidence artifact.
9. Append the artifact to the OASSE evidence chain.

A runnable launcher and HTML console were subsequently packaged around this validated path. OBS was the deployed NeoMundi wire mode. GOV remained the internal OASSE designation for the controlled governance scenario.

4. NeoMundi measurements and objects used

The articulation operationally consumed NeoMundi's ALLOW and FLAG measurement-layer classifications. It also preserved supporting measurement and evidence objects where returned, including trace identifiers, measurement and normalizer versions, confidence, G-Score, G-Final, V-Score, stability measurements, processing time, source-response digests, and raw response evidence.

Interpretation boundary

ALLOW and FLAG were treated as observational classifications, not authorization decisions.

5. Separation of measurement, interpretation, and action

Measurement. NeoMundi produced real-time measurements of AI behavior at runtime and returned measurement-layer classifications and supporting evidence.

Interpretation. The integration layer normalized selected NeoMundi measurements into versioned and traceable OASSE facts. OASSE FAM independently evaluated the original request under declared policy.

Decision and action. Gatekeeper composed the available results using the strictest applicable state and produced the final PROCEED, HOLD, or DENY action. OASSE then sealed and persisted the complete evidence record.

Joint architecture

NeoMundi measures. Oak & Sparrow determines and enforces.

NeoMundi × Oak & Sparrow | Measure. Determine. Enforce.

6. Observed experimental results

Focused 60-request experiment

The focused experiment completed 60 of 60 controlled cases with zero technical failures and 60 sealed evidence artifacts.

Layer	Outcome	Count
NeoMundi measurement layer	ALLOW	58
NeoMundi measurement layer	FLAG	2
OASSE FAM	ALLOW	50
OASSE FAM	BLOCK	10
Gatekeeper	PROCEED	48
Gatekeeper	HOLD	2
Gatekeeper	DENY	10

The cross-layer composition demonstrated both directions of authority. Ten NeoMundi ALLOW classifications did not override independent OASSE policy blocks and therefore became DENY. Two requests admissible under FAM received NeoMundi FLAG classifications and moved to HOLD.

All 60 NeoMundi trace identifiers were unique and reconciled one-to-one with NeoMundi audit evidence. All source-response digests were unique, and the complete 60-record evidence segment remained sequential and internally continuous.

Focused-run performance

Performance boundary	Observed result
Average full-pipeline duration	20.336 seconds
Average NeoMundi processing time	20.024 seconds
Average OASSE-plus-transport residual	311.74 milliseconds
Median residual	299.72 milliseconds
P95 residual	366.73 milliseconds
P99 residual	539.07 milliseconds
Maximum residual	776.39 milliseconds
Effective throughput	177.01 sealed artifacts per hour

The residual includes network transit, serialization, bridge normalization, deterministic FAM evaluation, Gatekeeper composition, sealing, persistence, and client overhead. It is not an isolated processor timer.

Controlled 1,000-request experiment

The broader benchmark attempted 1,000 calls. NeoMundi recorded all 1,000 measurements, OASSE sealed 996 local artifacts, and four calls ended in local 60-second response timeouts. The local sealing rate was 99.6 percent.

Layer	Outcome	Count / base
NeoMundi measurement layer	ALLOW	253 / 1,000
NeoMundi measurement layer	FLAG	747 / 1,000
OASSE FAM	ABSTAIN	115 / 996
OASSE FAM	ALLOW	232 / 996
OASSE FAM	BLOCK	440 / 996
OASSE FAM	REVIEW	209 / 996
Gatekeeper	PROCEED	47 / 996
Gatekeeper	HOLD	509 / 996
Gatekeeper	DENY	440 / 996

The four timed-out calls remained visible in the attempted population and were preserved as reliability evidence rather than rewritten as successes.

1,000-request performance

Performance boundary	Observed result
Average full-pipeline duration	20.414 seconds
Average NeoMundi processing time	20.023 seconds
Average OASSE-plus-transport residual	390.77 milliseconds
Median residual	381.79 milliseconds
P95 residual	482.00 milliseconds
P99 residual	701.57 milliseconds
Maximum residual	907.09 milliseconds
Effective throughput	174.26 sealed artifacts per hour

Post-FAM Gatekeeper finalization

The narrower post-FAM Gatekeeper finalization stage completed below 250 milliseconds for all 996 successfully sealed actions:

Finalization boundary	Observed result
Mean	0.0414 milliseconds
Median	0.0312 milliseconds
P95	0.0660 milliseconds
P99	0.1577 milliseconds
Maximum	1.9541 milliseconds

Performance interpretation

The combined architecture sustained deterministic composition, action generation, evidence sealing, and trace preservation across both focused and larger controlled populations.

7. New information or capability provided

OASSE classifies NeoMundi's contribution as complementary and operationally distinct.

OASSE already had deterministic policy determination, action selection, sealing, and evidence persistence. NeoMundi added an independent runtime measurement and traceability layer. The clearest example occurred in the focused experiment: two FAM-admissible requests moved to HOLD after NeoMundi FLAG classifications, while ten NeoMundi ALLOW classifications remained subject to stricter OASSE policy blocks.

The measurement layer affected operational routing without becoming the authorization authority.

8. Main operational value

The principal operational value was a complete, separable, and reconstructable decision path:

- what NeoMundi measured
- what OASSE policy determined
- what Gatekeeper did
- which evidence supported each stage
- how long each measurement boundary required

The experiment demonstrated a functioning measurement-to-policy-to-action-to-evidence pipeline with independent runtime measurement, deterministic interpretation, operational routing, traceable provenance, case-level audit reconstruction, chained evidence integrity, and measurable latency and throughput.

9. Main limitations and remaining uncertainties

The principal early difficulty was interface semantics. GOV was used as the internal OASSE scenario designation, while the deployed NeoMundi interface required OBS as the wire mode. The integration was corrected without changing the architecture.

Remaining limitations include:

- controlled rather than production populations
- the need for broader testing across models, domains, and operating conditions
- the need to identify which individual NeoMundi measurements remain consistently decision-relevant across policy environments
- the absence of an independent external ground truth for every behavioral measurement
- the need to compare the OASSE articulation with the other independent infrastructures before drawing a conclusion about general metrological applicability

10. Conclusion justified at this stage

The NeoMundi layer provided complementary and operationally distinct information in the OASSE use case. It improved specific operational functions involving routing, traceability, evidence quality, and auditability.

The articulation demonstrated more than API compatibility. NeoMundi measurements were incorporated into an independently designed deterministic architecture, affected final actions in the focused experiment, and remained separate from OASSE's interpretation and decision authority.

Research conclusion

The OASSE experiment provides a substantive indication in support of the common-metrological-layer hypothesis. It does not, by itself, establish that hypothesis across all heterogeneous AI infrastructures. That conclusion must be assessed through comparison with the remaining independent articulations.

Conditions of use and publication

These responses may be used under the name Oak & Sparrow Systems Enterprise, with Joshua Johosky identified as Founder, Chief Executive Officer, and Principal Architect, subject to OASSE review and approval of the complete passage concerning the NeoMundi × Oak & Sparrow articulation.

The final paper must preserve the distinction between NeoMundi measurement, OASSE interpretation and determination, and Gatekeeper action and evidence. Benchmark figures must retain their correct denominators and measurement boundaries.

OASSE requests review and validation of the complete passage, direct quotations, diagrams, benchmark figures and tables, descriptions of Gatekeeper's capabilities, and conclusions drawn specifically from this articulation.

Proprietary boundary

Participation in the paper does not authorize publication or distribution of OASSE source code, policy packs, ontology materials, proprietary schemas, implementation details, binaries, launcher packages, patent-sensitive materials, or confidential collaboration artifacts.

Evidence basis

Evidence population	Validated record
Focused benchmark	run-d86e7abe7d55 60 attempted 60 sealed
Controlled scale benchmark	run-1000-full-20260729-a 1,000 attempted 996 sealed
Validated architecture	NeoMundi OBS measurement path OASSE FAM deterministic evaluation Gatekeeper strictest-state composition and chained evidence